

Use Case Flow Specification

Community Lost & Found Matching Network

Use Case ID: UC-03

Use Case Name: Match Lost & Found Items

Primary Actor: Finder / Owner

Supporting System: Community Lost & Found Matching Network

1. Preconditions

1. At least one active **Lost Item** or **Found Item** report exists in the system.
2. The report contains required item information such as category, location and relevant item details.
3. Item tags such as category, colour, brand or keywords have been extracted and associated with the report.
4. The matching service is available and geographical location information is accessible.

2. Main Success Scenario

1. The system receives a newly created or updated Lost/Found Item report.
2. The system retrieves the item's category, colour, extracted tags and geographical location.
3. The system searches active reports of the opposite type within the configured geographical radius.
4. The system compares the reports based on item category, tags and colour.
5. The system calculates a match score for compatible Lost and Found reports.
6. The system identifies one or more reports whose match score satisfies the required matching criteria.
7. The system records the potential match and displays the match result to the relevant Finder / Owner.
8. The system sends a match notification to the potential owner.

3. Alternate Flow

A1 — No Compatible Match Found

1. At Step 5, if no active report satisfies the matching criteria, the system does not create a match.
2. The system records that no compatible match was found.
3. The Lost/Found report remains active in the system.
4. The system does not send a match notification.
5. The system may attempt matching again when a new compatible report is added or an existing report is updated.

4. Postconditions

Success

- A compatible Lost and Found report is recorded as a potential match.
- The match score/result is available to the relevant user.
- A match notification is dispatched to the potential owner.

Failure / Alternate Flow

- No incompatible report is marked as a match.
- The original Lost/Found report remains active for future matching.